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Abstract

Hypertension remains a major public health concern in the Philippines, contributing significantly to cardiovascular morbidity and mortality. This study develops HRP AI, a web-based decision-support application for predicting clinical hypertension risk (≥ 140/90 mmHg) among Filipino adults using calibrated and explainable machine learning.

Using the DOST-FNRI 2015 National Nutrition Survey (NNS), a final dataset of 151,189 adults was constructed, with 22,988 hypertensive cases corresponding to a 15.20% prevalence rate. Eight machine learning algorithms were evaluated across multiple imbalance-handling configurations through a Two-Stage Randomized Search. XGBoost with algorithmic Class Weights was selected as the final predictive engine because it provided the strongest clinical compromise between sensitivity, discrimina tion, and calibration compatibility. At the standard 0.50 decision threshold, the model achieved 76.87% Accuracy, 76.51% Recall, an F1-Score of 0.5014, and an AUC of 0.8469. Under the selected 0.35 screening threshold, its Recall increased to 87.88%.

To improve probability reliability, Venn-Abers calibration was applied, reducing the Expected Calibration Error (ECE) to 0.0035. The final prototype integrates cali brated risk scores, uncertainty bounds, SHAP and LIME explanations, and Wachter style counterfactuals optimized through bounded Brent's method. The resulting system demonstrates that non-linear machine learning models can be made more transparent, probability-aware, and suitable for exploratory hypertension screening.

Keywords: Hypertension, Machine Learning, XGBoost, Probability Calibration, Venn-Abers, Explainable AI, SHAP, LIME, Counterfactual Explanations

1 Introduction

1.1 Background of the Study

Hypertension, often referred to as the "silent killer" due to its asymptomatic nature, is a leading non-communicable disease and a principal contributor to cardiovascular morbidity and premature mortality worldwide. In the Philippine context, the burden re mains substantial. Recent survey cycles, specifically the 2018 National Nutrition Survey (NNS), report a prevalence of 19.2% among Filipino adults aged 20 to 59, escalating dramatically to 35.0% among the elderly. Despite a reported disease awareness rate of 67.8%, blood pressure control remains critically poor; only 75% of those aware are on treatment, and of those treated, only 27% have their blood pressure successfully con trolled [1]. This gap contributes significantly to heart disease and stroke, which remain the top causes of mortality in the country.

Advances in artificial intelligence (AI) and machine learning (ML) enable scalable risk stratification and targeted public-health screening by leveraging large, multimodal datasets. These techniques have been applied across settings to predict cardiovascu lar risk, specifically utilizing the National Nutrition Survey (NNS) provided by the DOST-FNRI [2], and successful deployments as accessible web applications have been demonstrated [3, 4].

Two practical concerns limit the clinical utility of many ML models: (1) model explainability, and (2) probability reliability. Explainable AI (XAI) methods such as SHAP and LIME increase transparency and clinician trust [5, 6]. Independently, probability calibration techniques ensure that a model's predicted probabilities reflect observed frequencies, a requirement when predictions are used in decision-making workflows.

Recent methodological work integrates these concerns through the Cal ibrated Explanations framework, which produces uncertainty-aware explanations and Wachter-style counterfactuals optimized for minimal clinical perturbation aligned with calibrated probabilities [7, 8].

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1.2 Statement of the Problem

Existing ML systems for hypertension risk prediction, such as those developed by Eldem [9] and Naik et al. [3], often prioritize discrimination metrics like accuracy and AUC. However, these systems possibly yield unreliable probability estimates for clinical application since they have not been calibrated nor even been tested with calibration metrics. Current systems aiming for reliable explainability typically focus on post hoc testing of calibration rather than employing formal post hoc calibration methods to actively correct probability distortion, as seen in recent studies using survey data [4, 10].

This omission means that many explainable AI outputs are derived from uncalibrated scores, which misrepresent model confidence and undermine clinical trustworthiness. The current ”standard” approach to XAI is not completely ideal due to its nature; ex plaining the raw output of a black-box model without ensuring its probabilistic reliability can lead to misleading interpretations [11]. Furthermore, XAI and other ML systems are not without flaws; an explanation that justifies an overconfident but incorrect pre diction provides a false sense of security to the clinician [7]. Specifically, a Machine Learning System that utilizes the Philippine NNS to produce discriminative hyperten sion risk predictions, actively corrects probability unreliability using formal calibration methods (Platt scaling, Isotonic Regression, Venn-Abers), and generates a calibrated risk score, uncertainty interval, and actionable, Wachter-style counterfactuals, has not been systematically implemented and evaluated.

1.3 Research Questions

This study aims to answer the following research questions:

1. How can machine learning models (specifically k-Nearest Neighbors, Naive Bayes, Logistic Regression, Random Forest, AdaBoost, XGBoost, LightGBM, and Cat Boost) be developed and evaluated for predicting hypertension risk using the

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Clinical and Health, Anthropometry, and Dietary Survey Components of the DOST-FNRI 2015 National Nutrition Survey (NNS) dataset?

2. How can a Two-Stage Rigorous Optimization strategy be implemented to enhance model performance, and how can the Calibrated Explanations framework be integrated to provide reliable probability estimates using Platt Scaling, Isotonic Regression, and Venn-Abers predictors?

3. How can Explainable Artificial Intelligence (specifically SHAP and LIME) and bounded Brent’s optimization be integrated to extract transparent clinical risk drivers and prescribe physiologically realistic, Wachter-style lifestyle counterfac tuals?

4. How can the predictive model, along with its interpretable feature importance, uncertainty intervals, and actionable, Wachter-style counterfactuals, be effectively presented in a web application prototype designed as a decision-support tool for potential users like clinicians or health-conscious individuals?

1.4 Objectives of the Study

The general objective of this study is to develop a web application for hypertension prediction that utilizes a machine learning model enhanced with Calibrated Explainable AI to predict hypertension risk.

1.4.1 Research Objectives

Specifically, this study aims to:

1. Data Acquisition: Acquire and secure access to the microdata of the 2015 National Nutrition Survey (NNS) from the DOST-FNRI, specifically the Clinical and Health, Anthropometry, and Individual Dietary components.

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2. Data Merging: Systematically merge the three distinct datasets using unique re spondent identifiers ('hhnum' and 'member code') to create a unified, multimodal feature set for each respondent.
3. Target Variable Engineering: Construct the binary ground-truth variable 'Hy pertension' by applying the clinical threshold of Systolic Blood Pressure ≥ 140 mmHg or Diastolic Blood Pressure ≥ 90 mmHg to the averaged blood pressure readings found in the Clinical dataset.
4. Data Cleaning: Perform rigorous data cleaning within the NNS framework by filtering out invalid entries, dropping non-relevant anthropometric groups (infants, children, pregnant/lactating women), and removing administrative variables irrel evant to clinical prediction.
5. Imputing Outliers: Identify and handle outliers in numerical features using statistical methods to ensure robust model training without discarding valuable clinical signals.
6. Imputing Missing Values: Optimize the K-Nearest Neighbors (KNN) imputation strategy by conducting a masking experiment on 1% of the dataset to determine the optimal number of neighbors (□□), standardizing numerical variables prior to imputation to ensuring distance metric validity.
7. Feature Engineering: Construct new, clinically relevant features such as Body Mass Index (BMI) and Waist-to-Hip Ratio (WHR) from anthropometric measure ments, and categorical lifestyle levels (e.g., Smoking Level, Alcohol Level) from raw survey questionnaire responses.
8. Feature Selection: Apply a strict collinearity filter, leakage guards, and corre lation analysis to select the most predictive variables and reduce dimensionality, thereby improving model efficiency and reducing noise.
9. Model Development: Train eight candidate supervised machine learning algo rithms, including k-Nearest Neighbors (kNN), Naive Bayes, Logistic Regression,

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Random Forest, AdaBoost, XGBoost, LightGBM, and CatBoost, to predict binary hypertension status.

10. Hyperparameter Optimization: Optimize the performance of each candidate model using a Two-Stage Cross-Validation (CV) Optimization strategy, uti lizing a broad Randomized Search to identify promising regions of the hyper parameter space, followed by Local Refinement to pinpoint the configuration yielding the highest F1-score and AUC.
11. Class Imbalance Handling: Evaluate and apply imbalance handling strategies, specifically Baseline, Class Weights, SMOTE, SMOTENC, SMOTE + Class Weights, and SMOTENC + Class Weights, to address the disparity between hypertensive and normotensive classes in the dataset.
12. Probability Calibration: Apply and evaluate three post-hoc calibration tech niques—Platt Scaling, Isotonic Regression, and Venn-

Abers predictors—to en sure the predicted risk scores accurately reflect true empirical probabilities.

13. Explainability Integration: Implement Explainable Artificial Intelligence (XAI) modules by integrating SHapley Additive exPlanations (SHAP) and Local Inter pretable Model-agnostic Explanations (LIME). This integration will generate both global feature attributions (via SHAP) to identify dataset-wide risk drivers, and local attribution scores (via SHAP and LIME) to quantify the contribution of specific clinical and dietary variables to individual risk predictions.

14. Counterfactual Generation: Implement the extraction of Wachter-style counter factual explanations by minimizing an objective function that balances decision boundary crossing with a variance-normalized proximity penalty. The target probability is set slightly below the selected screening threshold of 0.35, allowing the system to identify the minimal clinically realistic single-feature perturbations required to alter a high-risk prediction.

15. Evaluation: Assess the models using discrimination metrics (Accuracy, Recall, 5

Precision, F1-Score, AUC) and calibration metrics (Expected Calibration Error, Log Loss) to select the optimal final model.

1.4.2 System Objectives

Specifically, this study aims to:

1. Design and build a prototype web application with a tiered explanation interface to serve as a decision-support tool.
2. Implement an interface for a user to input their required health data, specifically clinical variables (e.g., age, sex, smoking history), anthropometric variables (e.g., BMI, Waist-to-Hip Ratio), and dietary variables (e.g., daily intake of nutrients and food groups).
3. Integrate the trained and calibrated model to generate a personalized hypertension risk prediction based on the user's input.
4. Present the calibrated risk score, uncertainty interval (Venn Abers predictors), and both global feature importance (SHAP Global) and local patient-specific (SHAP Local and LIME) breakdowns for users or clinicians seeking a detailed technical analysis.

1.5 Significance of the Study

This study is significant to the following:

1. For the Patient: The system provides personalized hypertension risk estimates, calibrated probabilities, and interpretable explanations that may guide preventive action and support self-management of their health.
2. For Doctors and Healthcare Practitioners: The calibrated outputs, uncertainty intervals, and explanation interface offer transparent insights that may assist in

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clinical assessment, patient counseling, and early intervention planning, ensuring that decisions are backed by reliable probability estimates.

3. For Government and Policy Makers: The prototype demonstrates how survey based predictive tools may support screening workflows and wellness initiatives using national-level data such as the NNS, potentially informing public health policies regarding hypertension screening programs.
4. For the Research Community: The study provides a reference implementation for integrating probability calibration with explanation methods, contributing to future work on interpretable and uncertainty-aware predictive models.

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1.6 Scope and Limitations

The scope of this study and its limitations are as follows:

1. This study utilizes only the Clinical and Health, Anthropometry, and Dietary components of the 2015 NNS microdata for model training and evaluation.
2. The study focuses on training multiple machine learning algorithms, specifically k Nearest Neighbors, Naive Bayes, Logistic Regression, Random Forest, AdaBoost, XGBoost, LightGBM, and CatBoost.
3. The study applies specific imbalance-handling strategies, namely Baseline, Class Weights, Synthetic Minority Over-sampling Technique (SMOTE), SMOTENC, SMOTE + Class Weights, and SMOTENC + Class Weights.
4. The study performs probability calibration using Platt scaling, Isotonic Regres sion, and Venn-Abers predictors.
5. The system integrates SHapley Additive exPlanations (SHAP) and Local Inter pretable Model-agnostic Explanations (LIME) within the Calibrated Explanations

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framework to produce uncertainty-aware explanations. Specifically, SHAP is uti lized for both global (population-level) and local (individual-level) feature attribu tions, while LIME provides accessible local explanations. These are implemented alongside Wachter-style counterfactuals optimized via bounded Brent’s method.
6. The web application developed is a prototype decision-support tool intended for demonstration and exploratory use; it is not intended to replace professional medical judgment.
7. Model performance may be affected by the limitations of NNS data, such as cross sectional design, measurement error, and reporting biases, particularly in dietary information, as recall bias is inherent in self-reported 24-hour food recall surveys.
8. Generalizability of the hypertension risk prediction model to populations outside the NNS sampling frame or to clinical environments may require further validation and calibration.

1.7 Assumptions

1. Measurements from the National Nutrition Survey (NNS) are assumed to be of adequate quality for hypertension risk modeling.
2. The selected variables from the clinical, anthropometric, and dietary components are assumed to capture meaningful predictors of hypertension risk.
3. Calibration subsets used for Platt scaling, Isotonic Regression, and Venn-Abers are assumed to contain sufficient samples to avoid severe overfitting and yield stable probability estimates.
4. Users of the web application prototype are assumed to provide inputs within realistic ranges consistent with NNS measurement standards (avoiding outliers), which will be enforced via input validation logic within the application.

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5. The average systolic and diastolic blood pressure measurements used in this study are assumed to be resting blood pressure readings, consistent with standard clinical practice guidelines for hypertension diagnosis.

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2 Review of Related Literature

This section provides a comprehensive critical review of the existing body of knowledge relevant to the development of an explainable and calibrated machine learning system for hypertension risk prediction. The discussion is thematically organized into five key areas: (1) the epidemiological landscape of hypertension, specifically within the Philip pine setting; (2) the evolution and application of machine learning methodologies in cardiovascular risk stratification, including modern optimization strategies; (3) the criti cal role of Explainable Artificial Intelligence (XAI) in fostering trust in clinical decision support systems, specifically distinguishing between interpretability and explainability; (4) the often-overlooked necessity of probability calibration in risk modeling; and (5) a synthesis of these domains to highlight the specific research gap addressed by this study, emphasizing the transition from arbitrary counterfactuals to mathematically rigorous, Wachter-style prescriptive analytics.

2.1 Hypertension: A Global and National Health Crisis

Hypertension, clinically defined as persistently elevated arterial blood pressure, is uni versally recognized as a primary driver of cardiovascular morbidity and premature mor tality. The World Health Organization (WHO) identifies it as a ”silent killer” due to its asymptomatic progression towards severe complications, including stroke, myocardial infarction, heart failure, and chronic kidney disease. The condition often remains un detected until significant organ damage has occurred, necessitating proactive screening measures.

In the context of the Philippines, the burden of hypertension is both substantial and escalating, posing a severe challenge to the public health infrastructure. Dela Rosa et al. [12] highlight that despite various public health initiatives, the prevalence of un controlled hypertension remains alarmingly high. Their review of national surveillance data indicates persistent gaps in disease awareness, treatment adherence, and control

rates among Filipinos. Specifically, lifestyle factors prevalent in the region, such as diets high in sodium and processed foods, coupled with increasing urbanization and sedentary behavior, exacerbate the risk profile of the population. This epidemiologi cal reality underscores the urgent need for scalable, accessible screening tools that can identify high-risk individuals early in the disease trajectory, potentially before clinical symptoms manifest.

2.2 Machine Learning in Cardiovascular Risk Prediction

The limitations of traditional linear scoring systems, such as the Framingham Risk Score, have catalyzed a paradigm shift towards Machine Learning (ML) methodologies. Unlike traditional statistical methods, which often assume linear relationships between risk factors and outcomes, ML algorithms excel at modeling the high-dimensional, non-linear interactions inherent in complex physiological data.

2.2.1 The Shift from Linear to Non-Linear Models and Advanced Optimization

Recent literature demonstrates a clear trend towards the use of ensemble learning tech niques for medical diagnosis, moving away from simple probabilistic models like Naive Bayes or standalone logistic regression models that may fail to capture complex biologi cal dependencies. Naik et al. [3] and Pandey et al. [4] discuss how Artificial Intelligence (AI) and Digital Twins are reshaping hypertension prediction. Their findings suggest that algorithms capable of capturing complex feature interactions, specifically Random Forest and Gradient Boosting Machines (GBMs), consistently outperform these simpler baselines in terms of

16 discrimination metrics like the Area Under the Receiver Operating Characteristic (AUC-ROC) curve.

However, maximizing the predictive capability of these complex ensembles requires rigorous hyperparameter tuning. Traditional exhaustive methods, such as Grid Search, often suffer from the ”curse of dimensionality” when applied to algorithms like XGBoost

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or CatBoost, which possess vast parameter spaces. Recent methodological literature, specifically the foundational work of Bergstra and Bengio [13], demonstrates that Ran domized Search is significantly more effective than Grid Search for high-dimensional optimization. By sampling the parameter space statistically, it can identify optimal hyperparameter regions with a fraction of the computational cost. Building upon this, modern implementations frequently adopt a Two-Stage approach: utilizing a broad randomized search for initial landscape exploration, followed by a concentrated local re finement around the best-performing candidates. This ensures a mathematically robust balance between exploration and exploitation.

2.2.2 Application on Survey Data and Class Imbalance

The utilization of national health surveys for training predictive models is a growing area of interest in health informatics. Elvas et al. [14] emphasize the utility of multimodal datasets that combine clinical history with anthropometric and lifestyle variables. In the Philippines, the National Nutrition Survey (NNS) provides a rich, albeit complex, data source. However, effectively utilizing such survey data requires robust preprocessing to handle challenges inherent to the methodology, such as missing values due to participant non-response, the high dimensionality of continuous dietary data, and severe class imbalance (where normotensive individuals significantly outnumber hypertensive ones). Addressing this class imbalance is critical; standard algorithms biased towards the majority class produce superficially high accuracy but clinically dangerous false negative rates. The machine learning literature advocates for two primary interventions to resolve this: algorithmic cost-sensitive learning and synthetic data generation. Ap plying Class Weights directly penalizes the misclassification of the minority class during optimization. Conversely, data-level interventions like the foundational Synthetic Mi nority Over-sampling Technique (SMOTE) [15], and its variant for mixed continuous and nominal data (SMOTENC), are frequently employed to expand the decision bound ary of the minority class without the severe overfitting associated with naive random

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oversampling. Addressing these structural challenges through rigorous engineering interventions—such as optimal K-Nearest Neighbors imputation, strict collinearity fil tering, and robust resampling protocols—is central to ensuring model validity before calibration.

2.3 The Explainability Challenge: Explainable AI (XAI)

65 While the predictive power of ”black-box” models such as Deep Neural Networks and XGBoost is undisputed, their opacity poses a significant barrier to clinical adoption. Mohapatra et al. [16] argue that in high-stakes medical environments, trust is contingent upon transparency. A clinician cannot ethically or practically act on a risk score without understanding the rationale behind the prediction, nor can they explain the risk to a patient to motivate lifestyle changes without intelligible insights. It is crucial to distinguish between interpretability, which

refers to the extent to which a cause and effect can be observed natively within a system (often inherent in simple models like Decision Trees), and explainability, which involves using external techniques to elucidate the internal mechanics of complex black-box models. This study specifically seeks to extract explainability from high-performance ensemble models to bridge the gap between algorithmic accuracy and user trust.

To bridge this gap, Altukhi et al. [6] and Sreeja et al. [5] detail the rise of model agnostic XAI methods. SHapley Additive exPlanations (SHAP), grounded in coopera tive game theory, has become the gold standard for quantifying global and local feature importance. It provides a consistent way to attribute a prediction's value to each fea ture, ensuring that the sum of feature contributions equals the model output. Similarly, Ribeiro et al. [17] introduced Local Interpretable Model-agnostic Explanations (LIME), which provides intuitive, perturbation-based explanations for individual instances by ap proximating the complex model with a simpler, interpretable one (like a linear model) locally around the prediction.

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Beyond static explanations, the demand for actionable guidance has led to the de velopment of counterfactual explanations. Wachter et al. [18] pioneered a framework emphasizing that explanations should not merely describe the current state, but prescribe the minimal perturbation required to alter an undesirable outcome. In medical domains, generating these ”Wachter-style” counterfactuals is vital because it shifts XAI from de scriptive analytics to prescriptive intervention, offering patients specific, realistic targets (e.g., reducing BMI by a precise, minimal amount) rather than abstract, unachievable metrics.

2.4 The Reliability Gap: Probability Calibration

A predictive model is considered calibrated if its predicted probabilities align with the true empirical frequency of events. For instance, if a model predicts a 30% risk of hypertension for a specific group of patients, exactly 30% of that group should truly be hypertensive in reality.

Lofstr ¨ om et al. [¨ 11, 8] identify a pervasive issue in modern ML: complex models like Deep Learning and Boosting ensembles tend to be poorly calibrated. Because these algorithms focus on minimizing classification error or maximizing the margin, they often output overconfident probability scores (pushing probabilities towards 0 or 1). This results in a high Expected Calibration Error (ECE). In a clinical setting, an overconfident false alarm can lead to over-treatment and patient anxiety, while an underconfident miss can lead to negligence and a lack of early intervention. Therefore, calibration is not merely a statistical nicety but a safety requirement for decision support systems. Techniques such as Platt Scaling (fitting a sigmoid function) and Isotonic Regression (fitting a stepwise non-decreasing function) are standard post-hoc methods to correct this, while recent advancements like Venn-Abers predictors offer rigorous probability intervals that further quantify the empirical uncertainty of the prediction.

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2.5 Synthesis and Research Gap

The intersection of these fields reveals a distinct research gap. While numerous studies, such as that by Eldem [9], apply ML to hypertension, and others like Niu et al. [19] apply XAI for transparency, there is a scarcity of systems that explicitly integrate probability calibration into the explanation pipeline, particularly for Philippine health data utilizing the National Nutrition Survey (NNS). Most existing systems explain the raw risk score, which poses a significant clinical danger: if the underlying model is uncalibrated, the XAI tool might convincingly explain a probability that is statistically incorrect. Furthermore, standard counterfactual generators often ignore the feasibility of the suggested changes, leading to mathematically optimal but clinically irrelevant recommendations.

This study addresses these gaps by adopting the Calibrated Explanations framework proposed by Lofstr ¨ om et al. [¨ 7], heavily augmented with Wachter-style counterfactual generation. By combining state-of-the-art classifiers—optimized via a rigorous Two Stage Search—with post-hoc calibration methods and uncertainty-aware explanations, this research aims to produce a tool that is not only discriminatively accurate but also probabilistically reliable and transparent. Crucially, by integrating bounded optimiza tion routines to ensure minimal clinical perturbation, this integration ensures that the explanations and counterfactuals provided to the user are mathematically grounded in corrected risk estimates and physiologically actionable, directly addressing the trust deficit in AI-driven healthcare.

3 Theoretical Framework

This section elucidates the theoretical foundations and operational principles of the core machine learning algorithms, sampling strategies, calibration methodologies, and explainability frameworks utilized in this research. It provides the mathematical and conceptual basis for the system's design.

3.1 Heart Disease and Hypertension Modeling

Hypertension is the pathological state of sustained elevated blood pressure. It is the single most significant risk factor for cardiovascular disease. In the context of predictive modeling, the task is formulated as a binary classification problem: mapping a vector of input features □□ (encompassing clinical, anthropometric, and dietary variables) to a target label □□ ∈ {0, 1}, where 1 indicates the presence of hypertension based on the thresholds defined by the 2020 Philippine Clinical Practice Guidelines (SBP ≥ 140 mmHg or DBP ≥ 90 mmHg) [1], as measured in the NNS clinical component [2].

The DOST-FNRI 2015 National Nutrition Survey (NNS) was selected as the pri mary data source due to the critical need for cultural and genetic relevance. Existing hypertension models trained on Western populations (e.g., Framingham) or homoge neous datasets often fail to generalize to the Filipino demographic due to distinct dietary patterns, such as high sodium and rice consumption, as well as specific genetic predis positions and environmental factors. The 2015 NNS provides the most comprehensive, nationally representative snapshot of these local determinants, ensuring the model cap tures the specific risk profile of the Philippine population.

3.2 Data Preprocessing

Raw survey data is rarely suitable for direct algorithmic ingestion. Rigorous prepro cessing is required to standardize inputs, mitigate biases inherent in data collection, and

prevent artificial performance inflation.

3.2.1 Target Definition and Leakage Guard

In predictive modeling, data leakage occurs when information from outside the training dataset is used to create the model, or when the model is inadvertently given access to the target variable during training. Since the binary 'Hypertension' target is mathemati cally derived from the raw Systolic Blood Pressure (SBP) and Diastolic Blood Pressure (DBP) readings, retaining these raw continuous variables in the feature set would result in severe target leakage. To ensure the model learns to predict risk from behavioral, an thropometric, and dietary factors rather than simply "re-identifying" the blood pressure measurements, a strict Leakage Guard is executed immediately after target creation to drop all direct and indirect BP-related variables.

3.2.2 Dimensionality Reduction via Collinearity Filtering

High dimensionality, particularly with highly correlated features, can destabilize ma chine learning models (multicollinearity) and dilute the interpretability of feature at tributions. To address this, a Collinearity Filter is applied to the feature space. Vari ables exhibiting a Pearson correlation coefficient of □□ > 0.70 with another feature are flagged, and the redundant feature is purged. This ensures that the model trains only on high-signal, independent predictors, optimizing both computational efficiency and explanation clarity.

3.2.3 Handling Missing Values

Real-world survey data frequently contains missing entries due to non-response or measurement errors. To preserve data integrity and maximize the utilization of available information, this study employs advanced imputation techniques rather than simple row deletion.

The quality of K-Nearest Neighbors (KNN) imputation is highly sensitive to the choice of □□ (the number of neighbors) and the scale of the features. Since KNN relies on Euclidean distance to find similar respondents, variables with large magnitudes (e.g., Annual Income or Total Energy Intake) can disproportionately influence the neighbor selection process. Therefore, Standardization of all numerical variables is a prerequisite step before imputation. Furthermore, to objectively determine the optimal □□, an artificial masking experiment is employed. This involves randomly hiding 1% of the observed (non-missing) values in the dataset to create a "masked" copy. The original values serve as the ground truth. The masked dataset is then imputed testing specific odd values for □□ ∈ {3, 5, 7, 9, 11}, and the imputed values are compared against the ground truth using Root Mean Squared Error (RMSE) to select the optimal □□.

• KNN Imputation: For continuous numerical features, missing values are es timated using the K-Nearest Neighbors (KNN) algorithm with the optimized □□. Unlike mean or median imputation which reduces variance, KNN imputation finds the most similar samples in the multi-dimensional feature space and imputes the missing value based on the median of these neighbors. This method preserves the local structure and correlations within the data.

• Mode Imputation: For categorical features, missing values are imputed using the most frequent category (mode) of the nearest neighbors to maintain the probability distribution of discrete variables.

3.2.4 Feature Scaling

Feature scaling normalizes the range of independent variables. This step is applied specifically to all continuous numerical features in the dataset (e.g., Age, BMI, Total Energy Intake). This is critical for distance-based algorithms where features with larger magnitudes would otherwise dominate the objective function.

• Standardization (Z-score Scaling): Rescales features to have a mean (□□) of 0 18
and standard deviation (□□) of 1. This technique assumes that the data follows a Gaussian distribution and is less affected by outliers than normalization.

□□□□□□□□ =□□ − □□
□□

3.2.5 Class Imbalance Handling

Medical datasets frequently exhibit class imbalance, where the number of healthy indi viduals (majority class) significantly exceeds those with the condition (minority class). Standard algorithms often bias towards the majority class to maximize accuracy. This study explores resampling to address this by evaluating six distinct configurations:

• Baseline: No resampling or weighting applied, serving as the empirical control.

• Class Weights (CW): This technique modifies the algorithm’s loss function to penalize the misclassification of the minority class more heavily than the majority class. It was selected as a cost-sensitive learning baseline because it addresses imbalance directly within the model optimization process without altering the original data distribution, preserving the integrity of the clinical features.

• SMOTE (Synthetic Minority Over-sampling Technique): Synthesizes new minority instances by interpolating between existing continuous samples in the feature space [15]. This expands the decision boundary of the minority class without overfitting as severely as random oversampling.

• SMOTENC (SMOTE for Nominal and Continuous): An extension of SMOTE specifically designed to handle datasets with a mix of continuous and categorical features without requiring independent preprocessing steps.

• Hybrid Approaches (SMOTE + CW and SMOTENC + CW): Evaluates whether combining synthetic generation with algorithmic cost-sensitive learning yields superior discrimination boundaries compared to utilizing either method in isolation.

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3.3 Machine Learning Algorithms

This study utilizes a diverse suite of supervised learning algorithms, ranging from interpretable baselines to complex ensembles. A comparative approach using eight distinct algorithms was chosen to establish a rigorous performance baseline and identify the optimal inductive bias for this specific dataset.

3.3.1 Linear and Instance-Based Models

Logistic Regression (LR) A foundational statistical model for binary classification, Logistic Regression estimates the probability that an instance belongs to a particular class using the logistic (sigmoid) function. For a feature vector □□ = (□□1, . . . , □□□□), the predicted probability is calculated as:

□□(□□ = 1|□□) =1
1 + □□−(□□0+□□1□□1+···+□□□□□□□□)

While mathematically a linear model, it serves as an essential, highly interpretable parametric baseline to contrast against the complex non-linear decision boundaries formed by the tree-based ensembles.

k-Nearest Neighbors (kNN) A non-parametric, instance-based learner, kNN classi fies a new point based on the majority vote of its □□ closest

neighbors in the feature space. Proximity is determined using a distance metric, predominantly the Euclidean distance for standardized continuous variables, defined between two points □□ and □□ as:
□□(□□, □□) =
vt∑□□ □□=1
(□□□□ − □□□□)2
This algorithm was chosen because it captures local similarities in patient profiles, effectively mimicking clinical case-based reasoning where a doctor might compare a patient to similar cases they have seen before. Figure 1 demonstrates this process: a
20
new data point (green circle) is classified based on whether the majority of its neighbors within a defined radius (dashed lines) belong to the ”blue square” class or the ”red triangle” class.

Figure 1: Example of k-Nearest Neighbors (kNN) classification. (Source: [20])
Naive Bayes (NB) A probabilistic classifier based on Bayes' Theorem, Naive Bayes assumes strong (naive) independence between features given the class label. While this assumption rarely holds in complex medical data, NB is computationally efficient. It was selected as a baseline model to justify the use of more complex algorithms; if a simple probabilistic model performs equivalently to an ensemble, the computational cost of the ensemble is unjustified. Furthermore, its probabilistic nature provides a useful reference point for assessing the calibration of more complex models.

3.3.2 Ensemble Methods

Ensemble methods combine multiple base estimators to improve generalizability and robustness over a single estimator.
Random Forest (RF) A bagging ensemble that constructs a multitude of decision trees at training time, Random Forest outputs the class selected by the majority of the trees. It was selected for its inherent resistance to overfitting via bagging and feature randomness. It serves as a strong benchmark for tree-based performance and is generally robust to noise, making it highly suitable for survey data. As depicted in Figure 2, the model trains multiple independent trees (Tree 1, Tree 2, etc.) on random subsets of
21
data, and their individual predictions are aggregated (via summation or majority vote) to produce the final result.
Figure 2: Diagram of a Random Forest (RF) model. (Source: [21])
Adaptive Boosting (AdaBoost) A boosting technique that trains learners sequentially, AdaBoost focuses subsequent learners on the instances that were misclassified by pre vious ones by increasing their weights. This iterative error correction converts a set of weak learners into a strong classifier. AdaBoost is included to test the efficacy of iterative bias reduction, which can be particularly effective for improving performance on hard-to-classify instances, such as borderline hypertension cases. Figure 3 illustrates this sequential process, where misclassified samples (shown as larger plus/minus signs) are given increasing weight in subsequent iterations to force the model to focus on them.
22

Figure 3: Conceptual model of Adaptive Boosting (AdaBoost). (Source: [22])
XGBoost (Extreme Gradient Boosting) An optimized distributed gradient boosting library, XGBoost builds trees sequentially to minimize a regularized objective function. It is renowned for its execution speed and performance in structured tabular data com petitions. XGBoost was included specifically for its regularization capabilities (L1/L2), which are critical for preventing overfitting on high-dimensional medical data like the NNS, representing the current industry standard for tabular data performance. Figure 4 shows the additive nature of the model, where multiple weak learners (trees) are combined, with each new tree attempting to correct the residual errors of the previous ensemble.
23
Figure 4: Simplified structure of an XGBoost model. (Source: [23])
LightGBM and CatBoost These are advanced variants of gradient boosting that offer specific advantages. LightGBM uses leaf-wise tree growth and histogram-based algo rithms, making it highly efficient for large datasets; it was chosen to ensure the system remains scalable. CatBoost implements ordered boosting and processes categorical fea tures natively. It was specifically selected to handle the numerous categorical variables in the NNS (e.g., region, smoking status) without the information loss or dimensionality explosion often caused by standard One-Hot Encoding [24, 25].
3.4 Two-Stage Rigorous Hyperparameter Optimization
To maximize the predictive performance of the selected algorithms while managing the computational demands of high-dimensional national survey data, the study aban dons simplistic exhaustive grid searches. Instead, it employs a Two-Stage Rigorous Optimization workflow informed by the statistical efficiency of randomized parame ter exploration [13]. This structured approach searches statistical distributions (e.g., uniform, log-uniform) rather than fixed discrete lists.
24
• Stage 1 (Broad Exploration): A Randomized Search is conducted across the entire allowable hyperparameter space of the chosen models. This stage uti lizes 5-fold cross-validation running for 25 proxy epochs. This relatively short epoch allowance acts as an early-stopping mechanism during the search, allowing the system to rapidly evaluate diverse configurations and discard mathematically unpromising hyperparameter regions without wasting computational resources.
• Stage 2 (Local Refinement): The top candidate configurations identified in Stage 1 are isolated and subjected to a highly concentrated local search. This phase narrows the boundary distributions around the successful parameters and runs for 80 proxy epochs. This ensures meticulous fine-tuning of delicate weights, learning rates, and tree depths within the most promising ”neighborhoods” of the parameter space.
• Final Training and Validation Evaluation: The single best configuration that survives local refinement is advanced to final training. The model is trained on the full training split for up to 300 epochs, utilizing early stopping mechanisms eval uated against the validation holdout. This ensures the model converges optimally while definitively preventing overfitting.
3.5 Model Evaluation Framework

To rigorously assess the performance of the predictive models and the reliability of their probability estimates, a comprehensive suite of evaluation metrics is employed.

3.5.1 Discrimination Metrics

These metrics evaluate the model's ability to correctly classify individuals as hyperten sive or normotensive.

• Accuracy: The proportion of true results (both true positives and true negatives) 25
among the total number of cases examined. While intuitive, it can be misleading in imbalanced datasets.

□□□□□□□□□□□□□□□□ =□□ □□ + □□ □□
□□ □□ + □□ □□ + □□□□ + □□□□

Where □□ □□ (True Positive) is the number of hypertensive patients correctly iden tified, □□ □□ (True Negative) is the number of healthy patients correctly identified, □□□□ (False Positive) is the number of healthy patients incorrectly identified as hypertensive, and □□□□ (False Negative) is the number of hypertensive patients incorrectly identified as healthy.

• Recall (Sensitivity): The proportion of actual hypertensive cases that are correctly identified by the model. In medical screening, high recall is critical to minimize false negatives (missed diagnoses).

□□□□□□□□□□□□ =□□ □□
□□ □□ + □□□□

• Precision: The proportion of predicted hypertensive cases that are actually posi tive. This metric is important for minimizing false alarms.

□□□□□□□□□□□□□□□□□□ =□□ □□
□□ □□ + □□□□

• F1-Score: The harmonic mean of Precision and Recall, providing a single metric that balances the two. It is particularly useful when the class distribution is uneven.

□□1 = 2 ×□□□□□□□□□□□□□□□□□□ × □□□□□□□□□□□□
□□□□□□□□□□□□□□□□□□ + □□□□□□□□□□□□

• Area Under the ROC Curve (AUC): Measures the ability of the model to dis tinguish between classes across all classification thresholds. An AUC of 0.5 represents random guessing, while 1.0 represents a perfect classifier. Values be tween 0.5 and 1.0 indicate that the model has some predictive power, with higher
26
values representing better discrimination capability.

3.5.2 Calibration Metrics

These metrics evaluate the reliability of the model's predicted probability scores.

• Log Loss (Cross-Entropy Loss): Penalizes confident but wrong predictions. A lower log loss indicates better calibrated probabilities.

□□□□□□□□□□□□□□ = −1□□∑□□ □□=1
[□□□□log(□□□□) + (1 − □□□□) log(1 − □□□□)]

Where □□ is the total number of samples, □□□□is the true label (0 or 1) of sample □□, and □□□□is the predicted probability that sample □□ belongs to the positive class.

• Expected Calibration Error (ECE): Measures the difference between predicted confidence and empirical accuracy. The predictions are grouped into bins, and the weighted average of the absolute difference between accuracy and confidence in each bin is calculated. To calculate ECE, the samples are partitioned into □□ equally spaced bins (e.g., 10 bins of

size 0.1). Let □□□□ denote the set of indices of samples falling into bin □□. The accuracy and confidence of bin □□ are defined as:
acc(□□□□) =1|□□□□ |∑ □□∈□□□□
1(□□□□ = □□^□□)
conf(□□□□) =1|□□□□ |∑ □□∈□□□□
□□^□□
The Expected Calibration Error is then calculated as:
ECE =
∑□□ □□=1
|□□□□ |
□□|acc(□□□□) − conf(□□□□)|
Lower ECE indicates a better-calibrated model. 27
3.6 Probability Calibration
31 44 87 33 7
Calibration addresses the reliability of the prediction. A model is calibrated if for all instances predicted with confidence □□, the actual probability of the class is □□. The inclusion of post-hoc calibration is the primary technical contribution of this study, addressing the reliability gap identified in the literature.
• Platt Scaling: A parametric method that fits a logistic regression model to the raw scores (logits) of the classifier. It assumes the distortion distribution is sigmoid. It produces a calibrated probability □□^ using the formula:
□□^(□□ = 1| □□ (□□)) =1
1 + exp(□□ □□ (□□) + □□)
Where parameters □□ (slope) and □□ (intercept) are scalar parameters learned by minimizing the negative log-likelihood (Log Loss) on the calibration set. This was selected as the standard benchmark for parametric calibration, necessary to correct the tendency of models like SVM and boosting ensembles to produce uncalibrated scores.
• Isotonic Regression: A non-parametric method that fits a piecewise constant non-decreasing function to the raw scores. It learns a function □□ that transforms uncalibrated scores □□□□to calibrated probabilities □□□□ by solving the following opti mization problem:
min□□
∑□□ □□=1
(□□□□ − □□(□□□□))2
Subject to the constraint that □□ must be monotonically non-decreasing (□□□□ ≤ □□ □□ ⟹ □□(□□□□) ≤ □□(□□ □□)). It is more flexible than Platt scaling and can correct non-sigmoid distortions, provided sufficient data is available. It serves as the standard benchmark for non-parametric calibration.
28
• Venn-Abers Predictors: A rigorous framework based on conformal prediction. Unlike point estimators, Venn-Abers produces a multi-probability prediction in terval [□□0, □□1]. For a test instance with score □□□□□□□□□□□, it fits two separate Isotonic Regressions (□□0 assuming class 0, and □□1 assuming class 1) on the calibration set combined with the test instance. This yields:
□□0 = □□0 (□□□□□□□□□□□)
□□1 = □□1 (□□□□□□□□□□□)
A single calibrated probability point estimate □□ is derived using the minimax
optimal combination:
□□ =□□1

1 − □□0 + □□1

This produces a probability interval that provides a more informative representa tion of predictive uncertainty, making it safer for medical decision-support settings than relying on a single uncalibrated point estimate.

3.7 Explainable Artificial Intelligence (XAI)

The Calibrated Explanations framework requires specific XAI methods that can operate on calibrated probabilities to ensure transparency.

3.7.1 SHAP (SHapley Additive exPlanations)

SHAP assigns each feature an importance value for a particular prediction [26]. Based on the concept of Shapley values from cooperative game theory, it calculates the marginal contribution of feature □□ across all possible feature subsets □□ (out of the total set of features □□):

29

□□□□ =
∑
□□⊆□□\{□□}
|□□|!(|□□| − |□□| − 1)!
|□□|![□□□□ (□□ ∪ {□□}) − □□□□ (□□)]

Where □□□□ (□□) is the model's prediction for instance □□ using only the subset of features □□. This mathematical foundation ensures that the feature attribution is fair and consistent. SHAP was selected because it is applied for both global attribution (to interpret overarching model behavior and dataset-wide risk drivers) and local attribution (to explain exactly why a specific patient was flagged as high risk). This capability is essential for personalized medicine and actionable insights.

3.7.2 LIME (Local Interpretable Model-agnostic Explanations)

LIME explains the predictions of any classifier by learning an interpretable model (such as linear regression) locally around the prediction. It perturbs the input data to probe how the model behaves in the immediate vicinity of the instance □□. Mathematically, LIME solves the following optimization problem:

explanation(□□) = arg min
□□∈□□L (□□ , □□, □□□□) + Ω(□□)

Where L is a loss function measuring how unfaithful □□ is to □□ in the locality defined by the proximity measure □□□□, and Ω(□□) is a penalty for the complexity of the explanation model [17]. LIME was chosen for its simplicity and model-agnostic nature. While SHAP is more theoretically consistent globally, LIME provides linear approximations that are often more intuitive for non-technical stakeholders (patients) to understand, serving as an effective first-glance explanation in the web application.

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3.7.3 The Calibrated Explanations Framework

This study integrates the above components using the framework proposed by [7]. Instead of explaining the raw output of a black-box model, this framework first calibrates the model and then generates explanations (feature weights and counterfactuals) based on the reliable, calibrated probabilities. This is the core novelty of the study: ensuring that the trust built by the explanation is grounded in a reliable risk estimate, preventing users from being misled by overconfident or underconfident predictions.

3.7.4 Counterfactual Explanations via Wachter-Style Bounded Optimization

The generation of counterfactual explanations in this study adopts the prescriptive prin ciples introduced by Wachter et al. [18]. Instead of

simply searching a predefined grid or optimizing for an arbitrary geometric distance, the system formulates counterfactual generation as a rigorous mathematical optimization problem. The goal is to find a counterfactual feature value □□′that safely crosses the selected screening boundary while minimizing the perturbation from the patient's original state □□.
For each actionable continuous feature, the system uses bounded Brent's method via scipy.optimize.minimize scalar to minimize the following objective function [27]:
□□(□□′) = □□ · (□□ (□□′) − □□□□□□□□□□□□□□□)2 +(□□′ − □□)2
□□2
Where:
• □□ (□□′) is the model's predicted probability at the proposed counterfactual value □□′.
• □□□□□□□□□□□□□□□ = 0.34 represents the target probability slightly below the selected 0.35 screening threshold. This acts as an anchor to ensure that the proposed change shifts the prediction below the deployed screening boundary rather than the default
31
0.50 classification boundary.
• □□ = 0.5 is the hyperparameter balancing the trade-off between reaching the exact target probability and keeping the feature change as minimal as possible. • The term (□□′−□□)2
□□2serves as a variance-normalized proximity penalty. Dividing by the feature's variance or scale width (□□2) ensures that features with large native
scales (e.g., caloric intake) are not unfairly penalized compared to features with small scales (e.g., Waist-to-Hip Ratio).
Unlike naive grid-sweep methods, Brent's optimization efficiently searches for the optimal single-feature change within the physiological bounds of that variable.
32
4 Design and Implementation
This section details the practical realization of the theoretical framework into a function ing software system. It delineates the system architecture, the specific composition of the dataset, the data engineering processes, the hardware infrastructure required for model development, and the software design patterns employed to build the web application prototype.
4.1 System Framework
The research implements a full-stack Data Science pipeline. The workflow is structured into sequential phases: Data Ingestion, Preprocessing, Model Training, Calibration, and Deployment. This flow ensures that raw survey data is systematically transformed into actionable clinical insights.
33
Figure 5: Methodological Workflow Diagram.
34
4.2 Data Source and Variable Definitions
4.2.1 The DOST-FNRI 2015 NNS Dataset
The study utilizes microdata from the 2015 National Nutrition Survey (NNS) conducted by the Department of Science and Technology - Food and Nutrition Research Institute (DOST-FNRI). To create a comprehensive

feature set for hypertension modeling, three distinct survey components were merged using unique respondent identifiers:

1. Clinical and Health Component: Provides the ground truth labels (blood pres sure measurements) and behavioral risk factors (smoking, alcohol consumption).
2. Anthropometric Component: Provides physical body measurements necessary for calculating indices like BMI, Waist, and Hip circumferences.
3. Individual Dietary Component: Provides granular data on daily food intake, aggregated into 27 specific food groups and macronutrient totals.

4.2.2 Data Variables

The final dataset consists of 53 variables: 52 predictors and 1 target variable. Table 1 enumerates these variables, their roles in the model, and their descriptions. Note that fe alcohol level and fe smoking level are engineered features derived from raw survey questionnaires to create cleaner categorical inputs.

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Table 1: Variables Table for Hypertension Prediction Model: Clinical, Anthropometric,
and Lifestyle Features

Variable Name	Role	Description
Hypertension	Target	Binary outcome (1: Hypertensive [≥ 140/90 mmHg], 0: Normal)
age	Feature	Age of Respondent (years)
sex	Feature	Biological Sex (1: Male, 2: Female)
ethnicity	Feature	Ethnicity Code (0: Non-IP, 1: IP, 2: 2/3 Filipino, 3: 1/2 Foreign)
pa met	Feature	Physical Activity (Metabolic Equivalent)
fbs	Feature	Fasting Blood Sugar (mg/dL)
chol	Feature	Total Cholesterol (mg/dL)
tri	Feature	Triglycerides (mg/dL)
hdl	Feature	High-Density Lipoprotein (mg/dL)
ldl	Feature	Low-Density Lipoprotein (mg/dL)
waist		

Feature
Waist circumference (cm)
hip
Feature
Hip circumference (cm)
BMI
Feature
Body Mass Index (□□□□/□□2)
fe alcohol level
Feature
0: Never, 1: Former, 2: Moderate, 3: Binge; derived from alcohol status and binge drinking variables.
fe smoking level
Feature
0: Never, 1: Former, 2: Occasional, 3: Daily; derived from smoking status and current smoking variables.

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Table 2: Variables Table for Hypertension Prediction Model: Dietary Food Groups

Variable Name	Role	Description
epwt fg1	Feature	Cereals and cereal products
epwt fg2	Feature	Rice and rice products
epwt fg3	Feature	Corn and corn products
epwt fg4	Feature	Other cereal products
epwt fg5	Feature	Starchy roots and tubers
epwt fg6	Feature	Sugar and syrups
epwt fg7	Feature	Dried beans, nuts, and seeds
epwt fg8	Feature	Vegetables
epwt fg9	Feature	Green leafy and yellow vegetables
epwt fg10	Feature	Other vegetables

epwt fg11
Feature
Fruits
epwt fg12
Feature
Vitamin C-rich fruits
epwt fg13
Feature
Other fruits
epwt fg14
Feature
Fish, meat, and poultry
epwt fg15
Feature
Fish and fish products
epwt fg16
Feature
Meat and meat products
epwt fg17
Feature
Poultry
epwt fg18
Feature
Eggs
epwt fg19
Feature
Milk and milk products
epwt fg20
Feature
Whole milk
epwt fg21
Feature
Other milk products
epwt fg23
Feature
Fats and oils
epwt fg24
Feature
Miscellaneous
epwt fg25
Feature
Beverages
epwt fg26
Feature
Condiments and spices
epwt fg27
Feature
Other miscellaneous items

37
Table 3: Variables Table for Hypertension Prediction Model: Nutrient Totals
Variable Name

Role
Description
Total Food epwt
Feature
Total food weight (g)
Total Ener
Feature
Total Energy (kcal)
Total Prot
Feature
Total Protein (g)
Total Calc
Feature
Total Calcium (mg)
Total Iron
Feature
Total Iron (mg)
Total VitA
Feature
Total Vitamin A (RE)
Total VitC
Feature
Total Vitamin C (mg)
Total Thia
Feature
Total Thiamin (mg)
Total Ribo
Feature
Total Riboflavin (mg)
Total Nia
Feature
Total Niacin (mg)
Total CHO
Feature
Total Carbohydrate (g)
Total Fat
Feature
Total Fat (g)

4.3 Data Preprocessing and Feature Engineering
This section details the rigorous data transformation pipeline applied to the raw NNS microdata to ensure data quality, relevance, and model readiness.
4.3.1 Data Merging Strategy
The study integrates three distinct datasets—Clinical, Anthropometric, and Dietary—to create a unified respondent profile. The merge was performed using a left join operation on the primary key formed by the unique household number (hhnum) and member code (member code). The Clinical dataset served as the left (base) table to ensure that only respondents with recorded physiological data were retained.
38
4.3.2 Data Cleaning and Filtering

To align the dataset with the study’s scope and clinical guidelines, specific filtering steps were applied:

• Dropping Invalid Target Entries: Any record missing values for both
17 Systolic Blood Pressure (SBP) and Diastolic Blood Pressure (DBP) was immediately dropped, as ground-truth hypertension status could not be established for these individuals.

• Exclusion of Non-Relevant Anthropometric Groups: The raw dataset included specific ”anthrogroups” that fall outside the scope of standard adult hypertension guidelines. The following groups were dropped:

– Infants and Children (0–120 months): 43,459 infants (0-59 mos) and 59,383 children (60-120 mos) were excluded as pediatric hypertension re quires distinct percentile-based guidelines not covered by the 140/90 mmHg adult threshold.

– Adolescents (121–228 months): 90,682 entries were initially flagged in the clinical component. The 2020 Philippine Clinical Practice Guidelines man date a switch from percentile-based pediatric diagnosis to fixed adult cutoffs (≥ 140/90 mmHg) starting at age 13 [1]. While it is technically possible to isolate individuals aged 13 and above by cross-referencing their exact chronological age from the anthropometry dataset post-merge, integrating this cohort would necessitate the development of two distinct machine learn ing pipelines—one utilizing percentile-based targets for early adolescents and another utilizing fixed thresholds for older adolescents and adults. To prevent excessive architectural complexity and maintain a highly optimized, singular focus on the adult population, this demographic was excluded en tirely.

– Pregnant and Lactating Women: 3,625 pregnant women and 10,904 lac 39 tating mothers were excluded due to confounding physiological factors (e.g., gestational hypertension) that require specialized medical management dis tinct from the general adult population.

• The final dataset retained only the Adults (19.08 years and above) group, resulting in a consolidated analytical dataset of exactly 151,189 rows for analysis.

4.3.3 Feature Engineering

New variables were constructed to capture clinically relevant risk factors:

• Target Variable (Hypertension): A binary target variable was engineered based on the 2020 Philippine Clinical Practice Guidelines. A respondent
58 is labeled as Hypertensive (1) if their average SBP ≥ 140 mmHg OR average DBP ≥ 90 mmHg; otherwise, they are labeled Normal (0).

50 • Body Mass Index (BMI): Calculated from weight and height measurements using the standard formula □□□□/□□2.

• Lifestyle Levels & Cascading Fallbacks: Raw questionnaire responses were aggregated into ordinal levels for Smoking (0-3) and Alcohol (0-3). To maximize data retention, the backend implementation utilizes a cascading fallback logic: if a direct aggregate code (e.g., smoke status) is missing or recorded as ”Not Ap plicable” (codes like 99, 888888), the system systematically infers the behavioral level from supporting raw columns (e.g., falling back to current smoking, then to ever smk) to construct a robust, unified categorical feature.

4.3.4 Handling Missing Values and Outliers

• Outlier Detection: Numerical features were screened for biologically implausible values (e.g., negative age, extreme caloric intake) which were treated as missing values to prevent model distortion.

• Standardization for Imputation: Prior to imputation, all numerical variables are standardized using a 'StandardScaler'. This step is essential because KNN relies on Euclidean distance, which is sensitive to scale; standardizing ensures that variables with large magnitudes (e.g., Annual Income) do not dominate the neighbor selection process.
• Optimization of Imputation Parameters: To objectively determine the opti mal number of neighbors (□□) for imputation, an artificial masking experiment is conducted. This involves randomly hiding 1% of the observed (non-missing) values in the dataset to create a "masked" copy. The original values serve as the ground truth. The masked dataset is then imputed testing specific odd values for □□ ∈ {3, 5, 7, 9, 11}, and the imputed values are compared against the ground truth using RMSE (for numerical variables) to select the □□ that yields the lowest error.
• KNN Imputation Application: Using the optimal □□ identified from the masking experiment, missing values are imputed. For numerical features, missing entries are filled with the median of the neighbors. For categorical features, the mode (most frequent value) of the neighbors is used.
4.3.5 Feature Selection
To reduce dimensionality and noise, redundant variables were removed:
• Identifiers: Administrative columns like province, region, and survey weights were dropped as they are not clinical predictors.
• Redundant Dietary Weights: Raw food weight variables (fg#) were dropped in favor of their edible portion weight equivalents (epwt fg#), which more accurately represent actual consumption.
• Origin Variables: Intermediate variables used solely for engineering the lifestyle levels (e.g., raw 'smoke status' strings) were removed to prevent information
41
leakage and multicollinearity.
4.4 Model Development and Optimization Strategy
The study employs a rigorous, multi-stage selection process to identify the optimal model.
4.4.1 Stage 1: Randomized Search and Hyperparameter Spaces
A Randomized Search with Stratified K-fold Cross-Validation was performed across the eight algorithms (XGBoost, CatBoost, LightGBM, Random Forest, AdaBoost, Logistic Regression, k-Nearest Neighbors, and Naive Bayes). Table 4 details the exact statistical distributions and parameter boundaries established for the optimization grid.
42
Table 4: Hyperparameter Search Space Distributions for Two-Stage Optimization
Algorithm
Hyperparameter
Search Space Boundary / Distribution
Logistic Regression
C (Inverse Regularization)
solver
Log-Uniform [0.0001, 100.0]
Categorical ['lbfgs', 'liblinear']
k-Nearest Neighbors
n neighbors
weights
metric

Integer [3, 30]
Categorical ['uniform', 'distance'] Categorical ['euclidean', 'manhattan']
Naive Bayes
var smoothing
Log-Uniform [1e-12, 1e-8]
AdaBoost
n estimators
learning rate
base depth
Integer [50, 300]
Uniform [0.01, 0.50]
Integer [1, 4]
Random Forest
max features
max depth
min samples split
min samples leaf
Uniform [0.2, 1.0]
Categorical [None, 10, 20, 30]
Integer [2, 19]
Integer [1, 9]
XGBoost
learning rate
max depth
subsample / colsample bytree reg lambda (L2)
Log-Uniform [0.01, 0.30]
Integer [3, 8]
Uniform [0.6, 1.0]
Log-Uniform [0.01, 10.0]
LightGBM
learning rate
max depth
num leaves
subsample / colsample bytree
Log-Uniform [0.01, 0.30]
Integer [3, 9]
Integer [15, 149]
Uniform [0.6, 1.0]
CatBoost
learning rate
depth
l2 leaf reg
random strength
Log-Uniform [0.01, 0.30]
Integer [4, 9]
Uniform [1.0, 10.0]
Uniform [0.1, 2.0]

• Configurations Tested: Each algorithm was strictly evaluated across six handling methodologies: (1) Baseline (No sampling), (2) Class Weights, (3) SMOTE, (4) SMOTENC, (5) SMOTE + Class Weights, and (6) SMOTENC + Class Weights.

• Selection Criteria: During Stage 1, models were allocated a limited epoch budget to act as an early-stopping mechanism. For every algorithm configuration, the top
43
models were extracted based on their F1-Score (to balance precision/recall) and AUC (to assess discrimination capability).

4.4.2 Stage 2: Secondary Selection for Calibration

From the pool of top models extracted in Stage 1, a concentrated local search and second filter was applied to select the candidates for calibration.

• Selection Criteria: The top performing models per algorithm type were selected based on Recall (sensitivity to hypertension cases) and Accuracy.

4.4.3 Stage 3: Probability Calibration and Final Selection

The selected models underwent post-hoc calibration using Platt Scaling, Isotonic Re gression, and Venn-Abers Predictors.

• Final Selection Criteria: The calibrated models were evaluated using a clini cally oriented hierarchy rather than a single metric alone. Recall was prioritized to preserve hypertension-positive case detection, AUC was used to verify discrim inative ability, and ECE and Log Loss were used to assess probability reliability after calibration. The final model was selected based on its overall suitability for screening deployment, balancing sensitivity, discrimination, calibration quality, and compatibility with the explainability pipeline.

4.4.4 Stage 4: Explainability and Counterfactual Generation

Following the selection of the final calibrated model, the explainability module is in stantiated to support the user interface outputs.

• Feature Masking for Actionability: To ensure that the generated counterfactuals are clinically realistic, a feature mask is defined. Variables such as Age, Sex, Eth
44
nicity, and Height are marked as Immutable (cannot be changed). Variables such as BMI, Dietary Intake, and Smoking/Alcohol levels are marked as Actionable.

• Counterfactual Search via Bounded Optimization: To ensure recommenda tions are both optimal and physiologically possible, the system utilizes bounded Brent's method (scipy.optimize.minimize scalar) to find the minimal per turbation of a given actionable feature that reduces the predicted risk below the selected 0.35 screening threshold. A variance-normalized penalty is applied to prioritize recommendations requiring the smallest absolute lifestyle change.

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4.5 System Design

4.5.1 Context Diagram

The system operates as a web-based application where the user provides health pa rameters, and the system returns a comprehensive risk assessment. The core process involves data standardization, model inference, probability calibration, and explanation generation. Specifically, the system receives Demographic Data, Clinical Data, and Dietary Intake directly from the user. It accesses an external Model Store contain ing the serialized ensemble model and Venn-Abers calibrator (persisted via a joblib pipeline). Finally, it outputs the Calibrated Risk Score (with its Uncertainty Interval), Tiered Explanations (LIME summary and SHAP waterfall), and prescriptive Actionable Counterfactuals back to the user.

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Figure 6: System Context Diagram.
4.5.2 Use-Case Diagram
Figure 7 illustrates the functional requirements of the system. The primary actor, the User (representing either a Patient or Clinician), interacts with the system to execute the following core functions:
• Input Health Data: Enter demographic, clinical, and dietary information. This process features an <<include>> relationship to Calculate Dietary Totals, rep resenting the dynamic auto-computation engine that reduces user friction.
• View Risk Assessment: Includes viewing the calibrated probability score along side its rigorous mathematical uncertainty interval.
• View Explanations: The user is provided a default, accessible LIME explanation. This use case contains an <<extend>> relationship to Request SHAP Explanation (Advanced), indicating that deeper global and local SHAP attributions are an optional, user-triggered flow.
46
• View Actionable Counterfactuals: Review the prescribed lifestyle and dietary modifications required to safely lower the predicted risk class.
Figure 7: System Use Case Diagram.
4.5.3 Activity Diagram
The operational flow is depicted in Figure 8, detailing both sequential processing and parallel explanation generation:
1. The user initiates the process by inputting their health data.
2. The system preprocesses the input through strict scaling and categorical encoding.
3. The backend ensemble model evaluates the data to generate an uncalibrated Raw Score.
4. The calibration layer applies the Venn-Abers transformation to the raw score to calculate the true probability estimate and uncertainty interval.
47
5. Parallel Fork: The system simultaneously (1) generates a local LIME explana tion and (2) executes bounded Brent's optimization to generate the Wachter-style counterfactuals.
6. Join: The parallel explanation processes consolidate, and the frontend renders the default Results Dashboard for the user.
7. Decision Node: The system checks if the user has requested the advanced view. If yes, it computes and displays the SHAP waterfall plot. If no, the process gracefully terminates.
Figure 8: System Activity Diagram.
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4.6 Technical Architecture
The system is built on a robust technical stack designed to support both the heavy computational load of model training and the lightweight requirements of the end-user web application.
8 4.6.1 System Architecture
The software architecture follows a modular design to ensure scalability and maintain ability:
8 • Backend: Developed in Python 3.12.3, leveraging libraries such as scikit-learn, PyTorch, and calibrated-explanations for the core logic.
• Frontend: Built using Streamlit, allowing for rapid prototyping of interactive data visualizations.

• Model Persistence: The trained pipelines (preprocessing + model + calibrator) are serialized using joblib for efficient loading during runtime.

4.6.2 Hardware Requirements for Model Training

The training phase involves computationally intensive tasks such as randomized search for hyperparameter optimization and the calibration of ensemble models. The recom mended specifications for the development environment are:

• Processor: Multi-core CPU with at least 8 cores to handle parallel processing tasks.

• Memory (RAM): Minimum of 16 GB to accommodate the in-memory processing of the large merged NNS dataset.

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• GPU: Dedicated GPU with CUDA support (e.g., NVIDIA RTX 3050 or higher) to accelerate the training of gradient boosting models (XGBoost, LightGBM, CatBoost) and parallelized ensembles utilizing cuML (Random Forest).

• Storage: SSD storage for fast I/O operations during data loading and model serialization, with at least 20 GB of free storage for imports and the dataset.

4.6.3 Requirements for Web Application Usage

The deployment environment is designed to be lightweight and accessible to users with standard consumer-grade hardware.

• Device: Any modern computer, tablet, or smartphone.

• Operating System: Platform-independent (Windows, macOS, Linux, Android, iOS).

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• Web Browser: A modern web browser with JavaScript enabled (e.g., Google Chrome, Mozilla Firefox, Microsoft Edge, Safari).

• Internet Connection: A stable internet connection is required to access the Streamlit application hosted on the server.

4.7 Project Development Timeline

The development of the hypertension prediction system is structured over a 24-week period, divided into five distinct phases ranging from data acquisition to final system evaluation.

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Table 5: Proposed Development Timeline

Phase

Activity

Duration

Description

Phase 1

Data Acquisition & Preprocessing

Weeks 1-4

Requesting NNS microdata, merging datasets, cleaning data (imputation, outlier removal), and engineering features.

Phase 2

Model Development & Optimization

Weeks 5-10 Implementing 8 algorithms, Randomized Search optimization, and evaluating imbalance handling configurations (Baseline, CW, SMOTE, SMOTENC, SMOTE+CW, SMOTENC+CW).

Phase 3

Calibration & XAI Integration

Weeks

11-15
Implementing Venn-Abers and Isotonic calibration, generating SHAP/LIME explanations, and validating ECE/Log Loss.
Phase 4
System
Development (Web App)
Weeks
16-20
Developing the Streamlit interface, integrating the backend model pipeline, and designing the tiered explanation UI.
Phase 5
Testing, Evaluation & Writing
Weeks
21-24
Conducting system testing, analyzing final results, and writing the manuscript.

Figure 9: Gantt Chart of Project Development Timeline.
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4.8 User Interface Design

This section presents the graphical user interface (GUI) of the web application, ar chitected specifically to maximize clinical usability, reduce cognitive load during data entry, and support mathematical transparency in its predictions

4.8.1 Landing Page

The Landing Page (Figure 10) serves as the entry point, establishing immediate trust and guiding the user to the assessment tool. It features a clean, medical-grade aesthetic with a prominent ”Start Risk Assessment” call-to-action. To contextualize the clinical importance of the tool, the application dynamically displays key statistics regarding the prevalence of hypertension (”The Silent Killer”) in the Philippines. The deployed prototype utilizes the empirical baseline prevalence of 15.20%, derived directly from the consolidated 2015 NNS adult dataset under the strict ≥ 140/90 mmHg clinical guidelines, anchoring the user’s expectations to true epidemiological data.

Figure 10: Web Application Landing Page Establishing Epidemiological Context.
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4.8.2 Progressive Input Interface and Dynamic Computations

To prevent user fatigue and ensure strict data integrity, the Input Interface is structured as a progressive, four-step form (Figures 11 through 14). The interface guides the user through logical clinical groupings: Clinical Profile, Anthropometrics, Lifestyle/Vices, and Dietary Intake.

Crucially, the Streamlit frontend utilizes a Dynamic Auto-Computation Engine to eliminate manual calculation errors. In the Anthropometric step (Figure 12), the system dynamically calculates the patient’s Body Mass Index (BMI) in real-time as height and weight are entered. Similarly, in the Dietary Intake module, the system auto-sums total caloric energy utilizing standard Atwater conversion factors (e.g., 4 kcal/g for carbohy drates and proteins, and 9 kcal/g for fats). This real-time mathematical synchronization ensures that the backend predictive engine receives a complete, internally consistent feature array without overwhelming the end-user.

Figure 11: Input Step 1: Foundational Clinical and Biological Profile.

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Figure 12: Input Step 2: Anthropometric Measurements with Dynamic BMI Calcula tion.

Figure 13: Input Step 3: Behavioral Risk Factors and Lifestyle Profiling.

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Figure 14: Input Step 4: Dietary Intake and Auto-Computed Macronutrient Totals.

4.8.3 Results Dashboard and Explainability Modules

Upon form submission, the system evaluates the patient data through the calibrated XGBoost pipeline and generates the Results Dashboard. This dashboard translates complex multidimensional predictions into a unified, highly interactive clinical interface, sequentially partitioned into descriptive, prescriptive, and explanatory modules.

The risk assessment output begins with the Calibrated Risk Score (Figure 15), dis played on a dynamic gauge anchored to the empirically selected 0.35 screening thresh old. This is immediately accompanied by the Venn-Abers Probability Interval, providing uncertainty bounds around the calibrated prediction. Following the diagnosis, the appli cation renders actionable Wachter-style Counterfactuals (Figure 16). Driven by bounded Brent's optimization, this module estimates the minimal feasible lifestyle modifications required to reduce the patient's modeled risk below the selected 0.35 screening threshold.

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Figure 15: Prediction Results: Calibrated Risk Gauge and Venn-Abers Uncertainty Bounds.

Figure 16: Prediction Results: Prescriptive Lifestyle Counterfactuals via Optimization.

Finally, to establish clinical trust, the dashboard deploys a Tri-Tiered Explainability framework. It first renders a mathematically exact Local SHAP Waterfall Plot (Figure 17) to explicitly quantify feature attributions. To aid in rapid visual interpretation, this is immediately paired with a Local LIME Divergent Bar Chart (Figure 18), allowing

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the clinician to establish visual "explanation consensus." The dashboard concludes with the Global SHAP Beeswarm Plot (Figure 19), contextualizing the patient's unique physiological drivers against dataset-wide trends.

Figure 17: Tri-Tiered Explainability (Part A): Local SHAP Waterfall Plot.

Figure 18: Tri-Tiered Explainability (Part B): Local LIME Bar Chart for Rapid Inter pretation.

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Figure 19: Tri-Tiered Explainability (Part C): Global SHAP Plot for Population Context.

58

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5 Results

10

This section presents the empirical findings of the study. It details the demographic distribution of the dataset, the performance metrics of the

machine learning models be fore and after calibration, the secondary empirical experiments used to justify sampling and threshold decisions,
96 the visual outputs from the Explainable Artificial Intelligence (XAI) techniques, and the final web application prototype.

5.1 Dataset Demographics and Class Distribution

24 Following the strict adherence to the adult diagnostic threshold of ≥ 140/90 mmHg, the final processed NNS dataset comprised 151,189 samples. The target variable exhibited a significant natural class imbalance, with 128,201 patients (84.80%) classified as normal or elevated (Class 0) and 22,988 patients (15.20%) classified as clinically hypertensive (Class 1).

Figure 20: Distribution of the Hypertension Target Variable (≥ 140/90 mmHg)

Furthermore, analyzing the continuous variables across these two classes reveals distinct clinical patterns. As illustrated in the feature distribution analysis, hypertensive individuals consistently exhibit
68 higher medians across key anthropometric markers such as Body Mass Index (BMI) and Waist Circumference.

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Figure 21: Violin Plots Illustrating Feature Distributions Across Classes

5.2 Performance of Pre-Calibrated Machine Learning Models

Following the execution of the Two-Stage Rigorous Optimization strategy, the first phase of evaluation focused on identifying the strongest-performing configuration for each algorithm based on discriminative performance. Table 6 presents the highest performing configurations observed across the evaluated algorithms, with XGBoost Class Weights included as an additional clinically important candidate because of its strong sensitivity under the screening-oriented threshold.

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Table 6: Top Performing Pre-Calibrated Models

Algorithm	Configuration	Accuracy	Recall	F1-Score	AUC
LightGBM	Class Weights (cw)	0.7428	0.8351	0.4968	0.8522
XGBoost	Base	0.8262	0.5125	0.4728	0.8519
Random Forest	Class Weights (cw)	0.7307	0.8553	0.4913	0.8487
CatBoost					

Class Weights (cw)
0.7300
0.8479
0.4885
0.8486
Logistic Regression
Base
0.8335
0.4449
0.4483
0.8486
XGBoost
Class Weights (cw)
0.7098
0.8788
0.4794
0.8469
AdaBoost
Base
0.7163
0.8671
0.4816
0.8456
k-Nearest Neighbors
Base
0.8247
0.4027
0.4112
0.8232
Naive Bayes
Base
0.7152
0.7949
0.4590
0.8094

The results demonstrate that LightGBM configured with Class Weights achieved the strongest overall discriminative performance, producing the highest AUC (0.8522) and the highest F1-Score (0.4968) among the evaluated models. Random Forest and CatBoost also demonstrated competitive sensitivity, both achieving Recall values above 84% while maintaining AUC values greater than 0.84.

However, the final model selection was not based on AUC alone. Since the system is intended for hypertension screening, the ability to identify positive hypertension cases was treated as a primary clinical requirement. A model with a marginally higher AUC but weaker deployment suitability would not automatically be preferred if it failed to provide a stable and clinically usable balance between discrimination, Recall, and calibration compatibility.

Within this context, XGBoost with Class Weights emerged as a strong clinical can didate. Although its AUC of 0.8469 was slightly lower than the highest-AUC models, it achieved very high Recall (0.8788) under the screening-oriented threshold used in the pre-calibration evaluation. This made it particularly relevant for the screening objective of the study, where false negatives are more clinically concerning than false positives.

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Therefore, XGBoost CW was retained as the primary candidate for post-calibration evaluation and final deployment consideration.

It is important to note that the pre-calibration classification metrics in Table 6 were computed using the decision threshold selected during the calibration-set threshold sweep. Therefore, the reported Accuracy, Recall, and F1-Score reflect the model's threshold-optimized screening behavior rather than the default 0.50 classification bound ary. For XGBoost CW, this lower operating threshold increased sensitivity, producing a Recall of 0.8788 at the cost of lower Accuracy.

5.3 Performance of Calibrated Machine Learning Models

Post-hoc probability calibration using Platt Scaling, Isotonic Regression, and Venn Abers Predictors was subsequently applied to the machine learning models. The objec tive of this phase was to evaluate whether probability correction could improve proba bilistic reliability while preserving clinically acceptable classification performance. Table 7 presents the five best-calibrated models ranked according to Expected Cali bration Error (ECE).

Table 7: Top 5 Best Calibrated Models Overall

Algorithm	Configuration	Calibration	Accuracy	Recall	AUC	ECE	Log Loss
AdaBoost	Class Weights (cw)	Venn-Abers	0.8480	0.0000	0.8226	0.0006	0.3236
AdaBoost	Class Weights (cw)	Isotonic Reg.	0.8480	0.0000	0.8226	0.0007	0.3236
Naive Bayes	Base	Isotonic Reg.	0.8480	0.0000	0.8091	0.0024	0.3411

XGBoost	Base	Base	0.8502	0.1190	0.8519	0.0027	0.3098
XGBoost	Class Weights (cw)	Venn-Abers	0.8484	0.0979	0.8469	0.0035	0.3140

The calibration phase revealed a significant trade-off between probability calibration and clinical sensitivity. Models that achieved extremely low ECE values frequently suf fered from severe reductions in Recall. Several of the best-calibrated models effectively
62
collapsed into majority-class prediction behavior, achieving near-perfect calibration while failing to identify positive hypertension cases.
This phenomenon demonstrates that minimizing calibration error alone is insufficient for selecting a clinically deployable hypertension screening model. A classifier that produces well-calibrated probabilities but fails to identify at-risk patients cannot function safely in a real-world medical setting. Consequently, model selection required balancing calibration quality with discrimination and minority-class sensitivity.
To identify the strongest clinically viable candidates, the best-performing uncalibrated baseline models were re-evaluated with emphasis on Recall, AUC, and calibration stability. Table 8 summarizes the strongest baseline candidates.

Table 8: Top Uncalibrated Candidate Models

Algorithm	Recall	Accuracy	AUC	Base ECE	Base Log Loss
LightGBM (cw)	0.8351	0.7428	0.8522	0.1999	0.4691
Random Forest (cw)	0.8553	0.7307	0.8487	0.1171	0.3691

CatBoost (cw)
0.8479
0.7300
0.8486
0.1877
0.4560
XGBoost (base)
0.5125
0.8262
0.8519
0.0027
0.3098
XGBoost (cw)
0.7651
0.7687
0.8469
0.1493
0.4118

The baseline analysis revealed two distinct model behaviors. LightGBM, Random Forest, CatBoost, and XGBoost CW prioritized minority-class sensitivity, producing substantially higher Recall values than their corresponding base configurations. How ever, these gains were accompanied by elevated calibration errors and increased Log Loss values. Conversely, the base XGBoost model demonstrated exceptional probabilistic stability, achieving the lowest ECE (0.0027) and lowest Log Loss (0.3098) among all baseline candidates while still maintaining highly competitive discriminative power (AUC = 0.8519). Nevertheless, its Recall of 51.25% remained substantially lower than the
63
clinically stronger class-weighted alternatives.
Among the clinically optimized models, XGBoost CW emerged as the strongest compromise between discrimination, sensitivity, and calibration stability. Although its ECE (0.1493) was higher than the naturally stable XGBoost base model, it remained substantially lower than the heavily overconfident LightGBM and CatBoost configu rations while preserving a clinically meaningful Recall of 76.51% at the default 0.50 boundary.
A distinction must be made between probability calibration and classification thresh olding. The row labeled Base in the post-calibration results does not apply a proba bility transformation; it preserves the original model probabilities. However, the post calibration evaluation computes threshold-dependent metrics using the default 0.50 decision boundary, whereas the pre-calibration evaluation reports metrics using the op timized threshold selected from the calibration split. This explains why XGBoost CW has identical AUC, Log Loss, and ECE across the base probability outputs, but different Accuracy, Recall, Precision, and F1-Score values. AUC, Log Loss, and ECE evalu ate the probability scores themselves, while Accuracy, Recall, Precision, and F1-Score depend on the chosen classification threshold.
Table 9: Effect of Decision Threshold on XGBoost CW Base Probability Outputs
Evaluation Setting
Threshold

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Accuracy
Recall
Precision
F1-Score
AUC
Log Loss
ECE
Threshold-Optimized Pre-Calibration Evaluation
0.35
0.7098
0.8788
0.3296
0.4794
0.8469
0.4118
0.1493
Default Boundary Base Evaluation
0.50
0.7687
0.7651
0.3729
0.5014
0.8469
0.4118
0.1493

Thus, the apparent discrepancy between the pre-calibrated and post-calibrated XG Boost CW results is not due to retraining or a change in probability outputs. It is caused by the use of different classification thresholds. The threshold-optimized evalu ation emphasizes screening sensitivity, while the default-boundary evaluation provides a standard reference point at 0.50. For final deployment, the study treats XGBoost CW as the predictive engine, Venn-Abers as the probability calibration layer, and the empirically selected 0.35 threshold as the screening operating point.
64
This trade-off established the core tension of the study: models optimized for sen sitivity tended to produce overconfident and poorly calibrated probabilities, whereas models with strong probabilistic reliability often sacrificed clinically important recall performance.
To further evaluate the balance between calibration quality and classification utility, the strongest candidate models were subjected to additional post-hoc calibration analysis. Table 10 summarizes the best-performing calibrator for each selected candidate model under the standard post-calibration evaluation setting.
Table 10: Final Calibration Shootout: Best Post-Calibrated Candidate Models

Algorithm Configuration Best Calibrator
Accuracy
Recall

Calib. ECE Calib. Log Loss
Combined Metric

LightGBM Class Weights
Venn-Abers
0.8506
0.1083
0.0036
0.3114
0.6610
CatBoost
Class Weights
Isotonic Reg.
0.8491
0.1173
0.0034
0.3127
0.6626
XGBoost
Class Weights Venn-Abers

0.8484
0.0979
0.0035
0.3140
0.6572

The final calibration analysis demonstrated that post-hoc calibration substantially reduced calibration error across the candidate models. ECE values were compressed to near-zero levels, and Log Loss values converged near 0.31. However, these improve ments in probability reliability also produced measurable reductions in Recall when evaluated at the default 0.50 boundary, demonstrating the practical trade-off between probability calibration and sensitivity preservation.

For this reason, the final architecture was not selected solely by the highest isolated post-calibration score. Instead, the final decision followed a clinically oriented hierarchy: (1) the model must preserve strong hypertension-positive detection before calibration, (2) it must retain competitive discriminative ability, (3) it must achieve acceptable post calibration probability reliability, and (4) it must remain suitable for integration with the calibrated explainability pipeline.

Under this selection framework, XGBoost with Class Weights calibrated using the Venn-Abers Predictor was selected as the final deployment architecture. XGBoost CW

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provided a strong pre-calibration clinical foundation by achieving high Recall while maintaining competitive AUC. After Venn-Abers calibration, it achieved near-zero cali bration error (ECE = 0.0035), indicating that its probability estimates were substantially corrected while preserving its role as a sensitivity-oriented screening model.

A critical distinction also emerged between the XGBoost Base and XGBoost CW configurations. The base XGBoost model demonstrated stronger native probability stability, achieving very low baseline ECE and Log Loss values. However, its lower Recall made it less appropriate as the final screening model. In contrast, the class weighted XGBoost configuration provided stronger minority-class detection, which better aligned with the clinical objective of identifying at-risk individuals.

Consequently, the study selected XGBoost CW combined with the Venn-Abers Pre dictor as the final model architecture. This choice reflects a clinically grounded com promise between sensitivity, discrimination, and calibrated probability reliability. The findings reinforce that model selection for medical screening should not rely on a single aggregate metric, but should instead jointly evaluate discrimination, calibration, and the clinical consequences of missed positive cases.

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5.4 Empirical Justification for Hyperparameter and Data Decisions

To ensure the integrity and clinical safety of the final pipeline, secondary empirical experiments were conducted to justify the selection of sampling techniques and decision thresholds.

5.4.1 Experiment G: Sampling Integrity and the Failure of Synthetic Oversam pling

Standard literature often relies on the Synthetic Minority Over-sampling Technique (SMOTE) and its categorical variant, SMOTENC, to handle class imbalance. A ded icated sampling experiment was conducted to compare the reliability of the baseline

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model, algorithmic Class Weights (cw), SMOTE, SMOTE with Class Weights, SMO TENC, and SMOTENC with Class Weights. This experiment evaluated whether syn thetic oversampling improved predictive performance and whether the generated syn thetic records remained clinically plausible. It must be noted that this experiment was conducted using a LightGBM classifier as the sampling test engine. LightGBM was selected for this sampling-integrity experiment because it was the fastest model to train among the high-performing gradient-boosting candidates, making it computationally practical for repeatedly evaluating multiple sam pling configurations. Therefore, the results in Table 11 should be interpreted as a con trolled sampling-integrity experiment rather than as the final XGBoost model-selection table.

Table 11: Experiment G: Predictive Performance Across Sampling Techniques (Light GBM)

Sampling Technique
Synthetic Rows
Impossible Synthetic % Accuracy

Recall
F1-Score
AUC
Baseline (No Sampling)
0
–
0.8516
0.1257
0.2049
0.8536
Class Weights (cw)
0
–
0.7295

0.8636
0.4926
0.8532
SMOTE
63,127
0.5497%
0.8418
0.3180
0.3793
0.8499
SMOTE + Class Weights
63,127
0.5497%
0.8418
0.3180
0.3793
0.8499
SMOTENC
63,127
0.5576%
0.8404
0.3263
0.3833
0.8492
SMOTENC + Class Weights
63,127
0.5576%
0.8404
0.3263
0.3833
0.8492

2 The results show that the baseline model achieved the highest AUC (0.8536) and highest Accuracy (0.8516), but its Recall was only 0.1257. This indicates that the base line model was overly conservative and failed to identify most hypertension-positive cases. In contrast, the 6 Class Weights configuration produced the highest Recall (0.8636) and highest F1-Score (0.4926), showing that cost-sensitive learning was the most effec tive strategy for recovering minority-class sensitivity without generating artificial patient records.
SMOTE and SMOTENC achieved competitive AUC values near 0.85, but their Recall values remained substantially lower than the Class Weights configuration. SMOTE
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achieved a Recall of only 0.3180, while SMOTENC achieved a Recall of 0.3263. Therefore, synthetic oversampling did not provide a clinically meaningful improvement in sensitivity compared with algorithmic Class Weights.
Beyond predictive metrics, the synthetic-data integrity check revealed that SMOTE and SMOTENC generated out-of-scope synthetic records. Both SMOTE-based ap proaches generated 63,127 synthetic rows. Among these, SMOTE and SMOTE + Class Weights generated 347 impossible synthetic records, equivalent to 0.5497% of their syn thetic samples. SMOTENC and

SMOTENC + Class Weights generated 352 impossible synthetic records, equivalent to 0.5576% of their synthetic samples.

Table 12: Examples of Out-of-Scope Synthetic Patients Generated by SMOTE

Synthetic Patient ID	Age	BMI	Total Energy	Clinical Viability Issue
Patient 91036	17.18	17.62	10445.09	Below adult cutoff / out-of-scope synthetic age
Patient 91279	17.06	17.16	10602.93	Below adult cutoff / out-of-scope synthetic age
Patient 91303	16.22	31.01	11074.60	Below adult cutoff / out-of-scope synthetic age
Patient 91412	14.70	32.52	9914.50	Below adult cutoff / out-of-scope synthetic age
Patient 91768	12.64	14.04	14906.39	Below adult cutoff / out-of-scope synthetic age

These examples show that ordinary SMOTE can interpolate across patient records in ways that produce structurally invalid synthetic observations. In particular, the generated records included ages below the intended adult modeling scope of the study. Since the target variable and preprocessing pipeline were designed around the adult hypertension threshold, the creation of below-cutoff synthetic observations weakens the clinical validity of the resampled training distribution. Although SMOTENC is designed to better preserve categorical feature structure, it still produced a comparable proportion of out-of-scope synthetic records in this experiment.

Consequently, synthetic oversampling was disqualified from the final pipeline. The experiment supports the use of algorithmic Class Weights because this approach im proved minority-class sensitivity without fabricating new patient records. Instead of altering the empirical feature space, Class Weights penalize minority-class misclassifi

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cation during model optimization, allowing the model to learn from authentic observed cases while avoiding synthetic artifacts. Since LightGBM was used primarily as the fastest high-performing test engine for evaluating sampling behavior, the conclusion drawn from Experiment G concerns the sampling strategy rather than the final classifier architecture.

5.4.2 Experiment H: Empirical Threshold Sweep

As previously demonstrated, relying on the default 0.50 decision threshold can produce insufficient sensitivity in imbalanced hypertension screening. To systematically examine the relationship between diagnostic sensitivity and false-positive control, an empirical threshold sweep was conducted across decision thresholds from 0.50 to 0.10.

This experiment was conducted using LightGBM as the threshold-sweep test engine because it was the fastest high-performing model to train and evaluate repeatedly. There fore, the results in Table 13 should be interpreted as an empirical threshold-selection experiment rather than as the final calibrated XGBoost deployment evaluation.

Table 13: Experiment H: Empirical Threshold Sweep Using LightGBM

Decision Threshold	Accuracy	Recall (Sensitivity)	Precision	F1-Score	Specificity
0.50 (Default)	0.8519	0.1379	0.5523	0.2207	0.9800
0.45	0.8493	0.2428	0.5094	0.3288	0.9581
0.40	0.8430	0.3863	0.4796	0.4280	0.9248
0.35 (Selected)	0.8280	0.5164	0.4437	0.4773	0.8839
0.30	0.8090	0.6343	0.4160	0.5025	0.8403
0.25	0.7860	0.7309			

0.3910
0.5094
0.7959
0.20
0.7569
0.8101
0.3650
0.5032
0.7473
0.15
0.7197
0.8740
0.3372
0.4867
0.6920
0.10
0.6694
0.9343
0.3070
0.4622
0.6219

The threshold sweep demonstrates the expected trade-off between sensitivity and 69
false-positive control. At the default 0.50 threshold, the model achieved the highest Accuracy (0.8519) and Specificity (0.9800), but its Recall was only 0.1379. This means that the model was highly conservative and failed to detect most hypertension-positive cases.
Lowering the threshold progressively increased Recall. At the most aggressive threshold of 0.10, Recall increased to 0.9343, meaning that the model detected the vast majority of positive cases. However, this came at the cost of substantially reduced Accuracy (0.6694), Precision (0.3070), and Specificity (0.6219), indicating that many normotensive individuals would be incorrectly flagged as high risk.
The threshold of 0.35 was selected as the operating point because it provided a more balanced clinical compromise. At this threshold, Recall increased to 0.5164 while Specificity remained relatively high at 0.8839. Although thresholds such as 0.30, 0.25, and 0.20 achieved higher Recall, they produced progressively larger reductions in Specificity and Precision. Thus, 0.35 was selected to improve sensitivity beyond the default threshold while avoiding an excessive false-positive burden.
Since LightGBM was used as the computationally efficient test engine for this exper iment, the threshold result should be interpreted as empirical evidence supporting the use of a lower operating threshold in imbalanced hypertension screening. The final de ployed XGBoost CW + Venn-Abers model follows the same clinical principle by using the selected 0.35 value as a screening threshold rather than as a diagnostic threshold.
5.5 Explainable Artificial Intelligence
With XGBoost CW + Venn-Abers established as the final calibrated model architecture, XAI techniques were applied to map both dataset-wide behaviors and individual patient level risk drivers.
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94 5.5.1 SHAP (SHapley Additive exPlanations)

Global feature attributions were extracted using SHAP. Figure 22 presents the mean absolute SHAP values, ranking the features by their average magnitude of impact on the model's output across the entire dataset. Age, Waist Circumference, and Body Mass Index (BMI) emerged as the highest-signal predictors for Hypertension (140/90).

Figure 22: Global SHAP Feature Importance (Mean Absolute SHAP Values)

To further illustrate the directional impact of these features, a SHAP Beeswarm plot (Figure 23) was generated. This plot reveals how higher values of specific features, such as Age and BMI, consistently push the model's prediction toward a positive hypertension

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classification.

Figure 23: SHAP Beeswarm Plot Demonstrating Feature Directionality and Density

Beyond global trends, SHAP was also utilized for local explanations. Figure 24 demonstrates a waterfall plot for an individual prediction, detailing exactly how the baseline expected probability was mathematically increased or decreased by the patient's specific physiological and dietary values to arrive at their final personalized risk score.

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Figure 24: Local SHAP Waterfall Plot for a Single Patient Prediction

While default SHAP waterfall plots provide rigorous mathematical breakdowns, their dense visualization can induce cognitive overload for non-expert users. To bridge this gap within the web application, the system extracts the local SHAP values and dynamically renders them as an accessible, divergent bar chart. This allows the user to immediately identify their top physiological risk contributors versus their protec tive factors, maintaining game-theoretic mathematical rigor while optimizing human readability.

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5.5.2 LIME (Local Interpretable Model-agnostic Explanations)

For a more accessible visualization of individual predictions within the web application, LIME was utilized. Figure 25 provides a straightforward, human-readable bar chart representing the LIME output, highlighting the top features supporting or contradicting the model's local decision boundary for a specific test instance.

Figure 25: Local LIME Explanation for a Single Patient Prediction

5.6 Web Application Prototype and Clinical Dashboard

The web application serves as the deployment vehicle for the trained XGBoost CW + Venn-Abers model, transforming calibrated risk estimates and explanation outputs into an interactive, highly visual decision-support interface.

5.6.1 Landing Page and Progressive Intake

The Landing Page serves as the entry point, establishing immediate clinical context. It features a clean, medical-grade aesthetic and dynamically displays the empirical baseline prevalence of 15.20% derived from the NNS dataset to anchor user expectations.

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Figure 26: HRP-AI Web Application Landing Page

Data entry is handled through a progressive, four-step intake form (Clinical Profile, Anthropometrics, Lifestyle, and Diet). This multi-step design prevents cognitive fatigue and includes dynamic computational safeguards, such as the automatic calculation of BMI and Total Caloric

Intake, ensuring the pipeline receives mathematically valid feature arrays.

Figure 27: Step 2 of the Progressive Intake Form featuring Dynamic BMI Calculation

5.6.2 Prediction and Results Dashboard

Upon submitting the clinical profile, the user is directed to the interactive Results Dashboard. This dashboard is strictly partitioned into three distinct analytical modules

to maximize explainability and clinical trust.

Module 1: Calibrated Risk Assessment. As shown in Figure 28, the system immediately displays the patient's exact calibrated risk score. Crucially, the interface visually enforces the empirically selected 35.00% screening threshold on a dynamic gauge. Furthermore, the dashboard actively renders the Lower and Upper Uncertainty Bounds generated by the Venn-Abers multipredictor, providing the clinician with a transparent interval of the prediction's reliability.

Figure 28: Module 1: Calibrated Risk Gauge and Venn-Abers Uncertainty Bounds

Module 2: Prescriptive Counterfactual Analysis. Transitioning from descriptive to prescriptive AI, the second module leverages bounded Brent's optimization to cal culate counterfactual interventions. The system explicitly anchors these calculations to the 0.35 screening boundary. For a patient designated as Low Risk, as depicted in Figure 29, the system correctly identifies that no immediate physiological perturbations are mathematically required to alter the classification, thereby confirming their healthy trajectory and preventing unnecessary medical anxiety.

Figure 29: Module 2: Prescriptive Analytics Confirming No Immediate Intervention Required

Module 3: Tri-Tiered Explainability. To reduce the black-box opacity of the prediction, the dashboard integrates three distinct explainability frameworks in a vertical flow. First, the application utilizes Local SHAP (Figure 30) to provide a game-theoretic mathematical breakdown of how the baseline probability was altered by the patient's specific traits.

Figure 30: Module 3 (Part A): Local SHAP Waterfall Plot for Exact Feature Attribution

Because dense waterfall plots can be difficult to interpret rapidly, the system imme diately follows this with a Local LIME analysis (Figure 31), rendered as an accessible, divergent bar chart. By providing two independent local explainers sequentially, the sys

tem allows clinicians to establish explanation consistency by visually verifying whether major features act in similar directions across explanation methods.

Figure 31: Module 3 (Part B): Local LIME Divergent Bar Chart for Rapid Visual Interpretation

Finally, the dashboard concludes with the Global SHAP Feature Importance beeswarm plot (Figure 32). This tri-tiered design ensures the clinician can cross-verify the patient's unique risk drivers while contextually anchoring those findings against the broader population-level baseline of the NNS dataset.

Figure 32: Module 3 (Part C): Global SHAP Beeswarm Plot for Population-Level Context

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6 Discussions

The development of the HRP-AI web application bridges the gap between computational risk stratification and interpretable clinical decision support. By integrating XGBoost with algorithmic Class Weights, Venn-Abers calibration, and Explainable Artificial Intelligence (XAI), this study provides a structured approach to hypertension screening using national-level Filipino health data. The empirical findings demonstrate that model selection in medical screening cannot rely on a single metric alone. Instead, predictive performance, minority-class sensitivity, probability reliability, threshold behavior, and explanation usability must be evaluated together.

The empirical results show that XGBoost with Class Weights provided the strongest overall clinical compromise for the final predictive engine. While other models achieved slightly higher isolated AUC or calibration scores, XGBoost CW offered a more appro priate balance for screening because it preserved clinically meaningful hypertension positive detection while maintaining competitive discrimination. The final selection of XGBoost CW combined with Venn-Abers calibration therefore reflects an end-to-end deployment decision rather than a simple ranking by one metric.

6.1 Clinical Hierarchy of Evaluation Metrics

When deploying machine learning within a healthcare ecosystem, the operational viabil ity of a predictive model cannot be judged purely on standard mathematical aggregates such as global accuracy. Instead, model selection must be dictated by the real-world clinical consequences of misclassification. Because this system is designed as a first-line screening tool for a chronic and often asymptomatic condition, the evaluation framework followed a clinically oriented hierarchy of metrics.

1. Primary Priority: Diagnostic Sensitivity (Recall). In medical screening, a false negative is the most dangerous error because it fails to identify a potentially

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